

Properties of Normal Distribution

Reading Objectives:

In this reading, you will understand the properties of normal and standard normal distributions, such as expectation and variance.

Main Reading Section:

Mean and Variance:

To compute $E[X]$, note that

$$E[X - \mu] = \frac{1}{\sqrt{2\pi} \sigma} \int_{-\infty}^{\infty} (x - \mu) e^{-(x-\mu)^2/2\sigma^2} dx$$

Letting $y = (x-\mu)/\sigma$ gives that

$$E[X - \mu] = \frac{\sigma}{\sqrt{2\pi}} \int_{-\infty}^{\infty} y e^{-y^2/2} dy$$

But

$$\int_{-\infty}^{\infty} y e^{-y^2/2} dy = 0$$

Implies $E[X - \mu] = 0$, or equivalently that $E[X] = \mu$.

Using this, we now compute $\text{Var}(X)$ as follows:

$$\begin{aligned} \text{Var}(X) &= E[(X - \mu)^2] = \frac{1}{\sqrt{2\pi} \sigma} \int_{-\infty}^{\infty} (x - \mu)^2 e^{-(x-\mu)^2/2\sigma^2} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \sigma^2 y^2 e^{-y^2/2} dy \end{aligned}$$

After some calculations, $\text{Var}(X) = \sigma^2$

Note: A very important property of normal random variables is that if X is normal with mean

μ and variance σ^2 , then for any constants a and b , $b \neq 0$, the random variable $Y = a + bX$ is also a normal random variable with parameters

$$E[Y] = E[a + bX] = a + bE[X] = a + b\mu$$

$$\text{Var}(Y) = \text{Var}(a + bX) = b^2 \text{Var}(X) = b^2 \sigma^2$$

Example 1: The power W dissipated in a resistor is proportional to the square of the voltage V . That is, $W = r V^2$

where r is a constant. If $r = 3$, and V can be assumed (to a very good approximation) to be a normal random variable with mean 6, and standard deviation 1, find $E[W]$ and $P\{W > 120\}$.

Solution: $E[W] = E[3V^2] = 3E[V^2] = 3(\text{Var}[V] + E^2[V]) = 3(1 + 36) = 111$

$$P\{W > 120\} = P(3V^2 > 120) = P(V > \sqrt{40}) = 0.37275.$$

Gaussian Mixture

A Gaussian Mixture is a function comprised of several Gaussians, each identified by $k \in \{1, \dots, K\}$, where K is the number of clusters of our dataset. Each Gaussian k in the mixture is comprised of the following parameters:

A mean μ that defines its center.

A covariance Σ that defines its width. This would be equivalent to the dimensions of an ellipsoid in a multivariate scenario.

A mixing probability π that defines how big or small the Gaussian function will be.

The Gaussian density function is given by:

$$N[x/\mu, \Sigma] = \frac{1}{(2\pi)^{D/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu)\right)$$

where x represents our data points, D is the number of dimensions of each data point. μ and Σ are the mean and covariance, respectively. The mixing coefficients are themselves probabilities and must meet this condition $\sum_1^K \pi_k = 1$.